

Varnika Singh

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Work rights: Full-time (Subclass 485, valid until Jan 2029) – no sponsorship required

SUMMARY

Graduate Software Engineer with experience building AI-powered applications and distributed systems across backend services, APIs and data pipelines. Skilled in developing LLM-based workflows, retrieval systems and scalable backend services using Python and modern web technologies. Strong interest in building real-world products, integrating systems end-to-end and shipping production-ready features.

KEY SKILLS

Languages: JavaScript, TypeScript, Python, R, SQL
Frontend & backend: React, Node.js, FastAPI, REST APIs, Redis, Kafka, Druid
Cloud & MLOps: AWS (Lambda, S3, API Gateway, Bedrock), Azure, Kubernetes, Docker, Terraform
AI / ML: NLP, LLMs, Prompt Engineering, RAG, PyTorch, TensorFlow, Hugging Face, scikit-learn
Tools: Jira, Confluence, Power BI, Retool, Copilot, CI/CD, Jenkins, Grafana, Prometheus, GitHub Actions

EDUCATION

Master of Computer Science – Machine Learning, Big Data, Cybersecurity Feb 2024 – Dec 2025
The University of Wollongong, NSW
Bachelor of Technology, Computer Science and Engineering – Internet of Things Jul 2017 – May 2021
Mody University of Science and Technology, India

WORK EXPERIENCE

Graduate Engineer Jun 2026 – Present
Helia Insurance, North Sydney, NSW

- Design, develop and deploy cloud-native applications and backend services using AWS.
- Develop and maintain APIs, cloud infrastructure and data solutions.
- Collaborate with engineering teams to deliver software to production using modern development practices, including CI/CD, testing and code reviews.

Software Engineer (Paid Contract) May 2026 – Jun 2026
Visagio, Sydney, NSW

- Extended a .NET production platform with multi-format data ingestion pipelines to reduce manual processing overhead.
- Designed and delivered a Major Events Calendar module within an Azure SQL and .NET enterprise environment.
- Delivered Phase 2 enhancements across a .NET, Snowflake and Power BI stack in a client-facing consulting engagement.

AI Intern (Paid Contract) Jul 2025 – Dec 2025
Hysata Pty. Ltd., Wollongong, NSW

- Built LLM-based product workflows using Python and AWS Bedrock, processing unstructured data into structured outputs across multiple systems.
- Designed retrieval pipelines (RAG, embeddings) to improve system accuracy and support real-world decision-making.
- Developed backend services and integrations using FastAPI, Snowflake and REST APIs to connect multiple systems.
- Built internal product features using Retool UI, enabling validation and interaction with AI-generated outputs.

Full Stack Developer May 2023 – Sep 2023
Varyra Enterprises (E-commerce store), India

- Built backend integrations using Shopify APIs and Node.js-style workflows, connecting logistics systems (FedEx, DHL, AusPost).
- Developed API-driven automation workflows, improving system reliability and reducing manual errors.
- Optimised frontend performance and system scripts to improve load time and user experience.
- Implemented real-time notifications using API integrations (SMS, email systems).

Web Developer Feb 2023 – May 2023
Shiva Shambu Nidhi Limited (NBFC), India

- Developed and stress-tested a secure web-based scheduling solution, ensuring 87% system uptime through continuous monitoring and manual verification.
- Collaborated on UI/UX responsiveness testing so automated scheduling flows were intuitive and error-free for end-users.

Cloud Developer Jul 2022 – Dec 2022
Hewlett Packard Enterprise (HPE), India

- Developed backend microservices using AWS (Lambda, API Gateway) and Python, supporting distributed system workflows.
- Built data pipelines using Kafka and SQL-based systems, enabling real-time data processing.
- Worked on CI/CD pipelines using Jenkins and Docker, improving deployment workflows.

System Software Engineer <i>Hewlett Packard Enterprise (HPE), India</i> - Designed and deployed Kubernetes-based systems, improving deployment efficiency by 40%. - Automated onboarding workflows, reducing manual intervention and increasing consistency across environments. - Optimised Jenkins pipelines to improve system performance and delivery timelines.	Aug 2021 – Jul 2022
Research and Development Intern <i>Hewlett Packard Enterprise (HPE), India</i> - Developed dynamic server profiles, enhancing data centre efficiency by 95% as part of the Digital Transformation initiative. - Built a Python-based IP retrieval framework using Redis, achieving 97.6% endpoint reach accuracy.	Feb 2021 – Jul 2021
AI Intern <i>Capgemini Engineering (Altran), India</i> - Leveraged Hugging Face transformers for text processing and model experimentation. - Developed a TensorFlow-based face recognition model (96% accuracy). - Improved model performance by 20% through OpenCV preprocessing and scikit-learn tuning.	Jun 2019 – Jul 2019

VOLUNTEERING

Volunteer <i>UOW Pulse Ltd., Wollongong, NSW</i> - Contributed to campus events such as Art Week and Unibar Trivia through logistics, crowd control and retail operations. - Recognised with the Events Engagement Award 2024 and named Volunteer of the Month (May) after 130+ hours of service.	Mar 2024 – Jul 2025
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PROJECTS

Privacy-First Intelligent Automation Engine <i>Independent project</i> - Developed an AI system for automated text processing and response generation using LLMs, handling high-volume user queries. - Designed retrieval-based workflows to improve response accuracy and contextual relevance. - Evaluated performance, balancing accuracy, latency and cost of inference.	Jan 2026 – Present
Capstone – NutriSnap (AI Food Recognition & Nutrition Tracking) <i>University of Wollongong, NSW</i> - AI-powered platform that estimates calories and nutrients from meal photos and logs daily intake. - Designed and deployed an end-to-end AI pipeline integrating computer vision models with backend APIs and database systems. - Ran experiments to compare model performance; implemented validation and error handling for data accuracy and reliability.	Feb 2025 – Oct 2025

CERTIFICATIONS

- Anthropic Academy – Claude AI Engineering (2026): Introduction to MCP; Introduction to Agent Skills; Building with the Claude API; Claude Code in Action - Google Cybersecurity Certificate - Microsoft Azure Fundamentals AZ-900 - IBM Security Intelligence and Event Management Mastery Award	
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REFERENCES

Available upon request